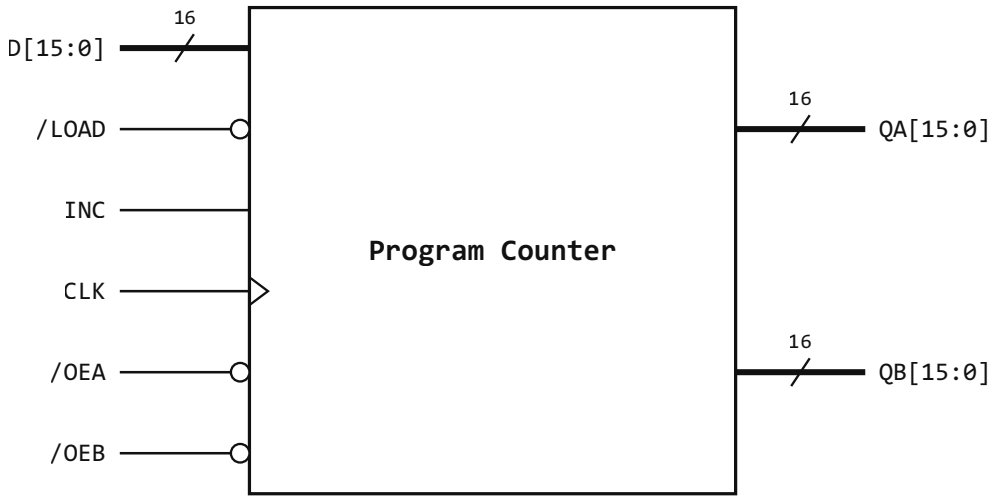
PC

Program Counter

16-bit register that can count by two. On the rising edge of CLK it loads D[15:0] when /LOAD is low; otherwise it adds 2 when INC is high (bit 0 unchanged); otherwise it holds. QA drives while /OEA is low, QB while /OEB is low; outputs are independent of the clock.

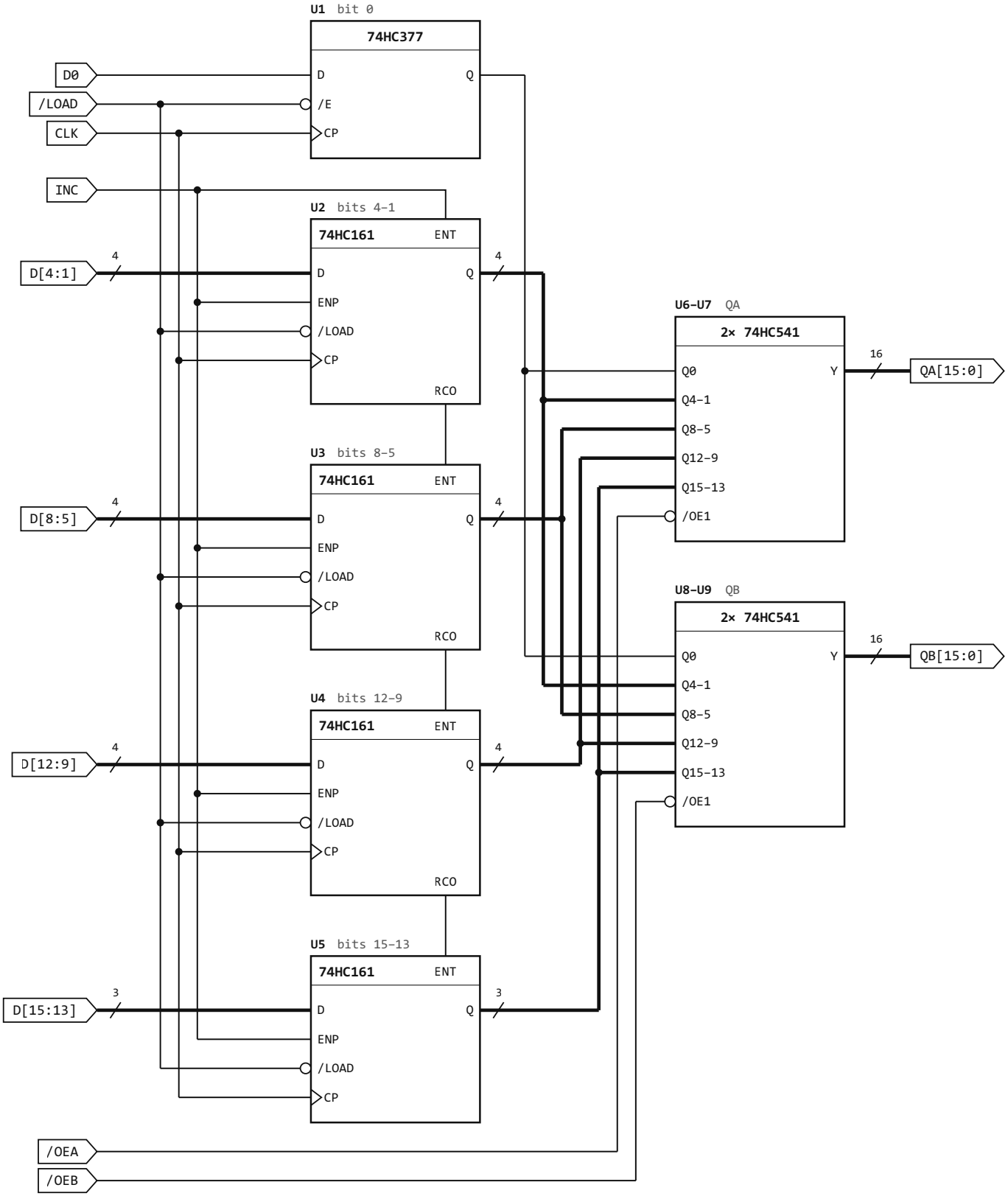
Width	16 bits	**Logic family**	74HC, 5 V	**Clocking**	Rising edge, synchronous
Count step	+2 (bits 15–1)	**Read ports**	2 × tri-state	**Chips**	9

1 PINOUT



Pin	Dir	Width	Active	Function
D[15:0]	In	16	High	Address to load (jump target)
/LOAD	In	1	Low	Load enable, sampled on CLK ↑; overrides INC
INC	In	1	High	Count enable: add 2 on CLK ↑
CLK	In	1	Rising ↑	Clock, rising edge
/OEA	In	1	Low	Enable read port A
/OEB	In	1	Low	Enable read port B
QA[15:0]	Out	16	High	Read port A, tri-state
QB[15:0]	Out	16	High	Read port B, tri-state

2 SCHEMATIC


U1 bit 0
74HC377
D
Q
/E
CP
D0
/LOAD
CLK
INC
U2 bits 4–1
74HC161
ENT
D
Q
ENP
/LOAD
CP
RCO
D[4:1]
4
U3 bits 8–5
D[8:5]
U4 bits 12–9
D[12:9]
U5 bits 15–13
D[15:13]
3
U6–U7 QA
2× 74HC541
Q0
Y
Q4–1
Q8–5
Q12–9
Q15–13
/OE1
16
QA[15:0]
U8–U9 QB
QB[15:0]
/OEA
/OEB

3 TRANSITION TABLES

Count / load

CLK	/LOAD	INC	PC next
↑	0	X	D
↑	1	1	PC + 2
↑	1	0	PC
0 / 1 / ↓	X	X	PC

↑ rising edge · X don't care · /LOAD overrides INC

Read ports

/OEA	QA[15:0]	/OEB	QB[15:0]
0	PC	0	PC
1	Z	1	Z

independent of CLK · Z high-impedance

4 BILL OF MATERIALS

Ref	Qty	Part	Description	Package
U1	1	74HC377	Octal D flip-flop with enable (1 bit used)	DIP-20
U2–U5	4	74HC161	4-bit synchronous binary counter	DIP-16
U6–U9	4	74HC541	Octal tri-state buffer	DIP-20
C1–C9	9	100 nF	Ceramic decoupling, one per chip	2.54 mm
Total	**18**			